



Acetyl 11-Keto Beta-Boswellic Acid Improves Neurological Functions in a Mouse Model of Multiple Sclerosis

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Abstract

Acetyl-11-keto- β -boswellic acid is one of the main active components of *Boswellia sp.* resin with the most potent anti-inflammatory activity. In recent years, herbal therapy has received considerable attention for the treatments of inflammatory and demyelinating diseases such as Multiple sclerosis (MS). Studies have shown that herbal compounds could enhance myelin repair and suppress inflammation. This study was designed to investigate the therapeutic effects of intraperitoneal administration of AKBA in Experimental Autoimmune Encephalomyelitis (EAE), as an animal model of MS. Following EAE induction in female C57BL/6J mice, animals were treated with AKBA and the levels of different serum inflammatory mediators, as well as motor functions, myelination, and inflammatory cell infiltration were assessed. Our results revealed that the application of AKBA alleviated EAE clinical severity, and suppressed inflammation, demyelination, leukocyte infiltration, and gliosis in EAE mice. Our findings suggest that the therapeutic effects of AKBA are likely a consequence of its neuroprotective and anti-inflammatory properties. The beneficial effects of AKBA may therefore provide new insights in various neuroinflammatory diseases such as MS and thereby could serve as a potential treatment candidate.

Keywords Acetyl 11-keto beta-Boswellic Acid · Multiple Sclerosis · Myelination · Neuroinflammation

Introduction

Multiple sclerosis (MS), an autoimmune disease of the central nervous system (CNS), is characterized by inflammation and demyelination [1, 2]. Key pathological features include focal infiltration of immune cells into the CNS parenchyma,

breakdown of myelin sheaths, astrogliosis, microglia activation, and neurodegeneration in the CNS of patients with multiple sclerosis [3–5].

During the early stages of MS, activated innate immune cells and the secreted cytokines promote the differentiation of T helper (Th) 1 and Th17 cells. These immune factors lead to inflammation and trigger a cascade of signaling resulting in demyelination and the death of oligodendrocytes. These cellular and molecular events ultimately lead to irreversible axonal loss and long-term disability of the afflicted individuals [6, 7]. Differentiation of native T-helper cells into Th1 and Th17 is induced by various chemokines and cytokines such as tumor necrosis factor (TNF)- α , interleukin (IL)-1, IL-6, IL-10, IL-12, macrophage inflammatory protein (MIP)-1, monocyte chemotactic protein (MCP)-1, prostaglandin (PG) E2, and interferon (IFN)- γ [8, 9].

Several cytokines and other molecules play important roles in the pathogenesis of MS. IFN- γ is a pro-inflammatory cytokine produced by activated T cells and NK cells. In MS, it helps activate immune cells and increases the production of other pro-inflammatory cytokines, fueling inflammation and demyelination in the disease [10]. TNF- α is another

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pro-inflammatory cytokine involved in MS that helps recruit immune cells to the CNS and contributes to myelin destruction. Blocking TNF- α has been shown to reduce disease activity [11]. IL-2 regulates immune responses by activating and amplifying autoreactive T cells, driving the autoimmune attack on myelin. Modulating IL-2 signaling may help treat MS [12]. IL-10 is an anti-inflammatory cytokine with immunosuppressive effects that can limit inflammation and the production of pro-inflammatory cytokines, potentially protecting against MS severity [13]. IL-17A is produced by Th17 cells and promotes CNS inflammation and myelin damage. Targeting IL-17A signaling may offer a treatment approach [13]. IL-6 activates immune cells and increases other pro-inflammatory cytokines, and higher levels have been detected in MS patient cerebrospinal fluid [14]. IL-1 β and IL-1 α fuel inflammation by activating immune cells and boosting pro-inflammatory molecule production, worsening the disease. PGE2 has both pro- and anti-inflammatory properties that influence the balance between the two responses in MS while modulating immune cell function. The chemokines MIP-1 and MCP-1 help recruit and activate immune cells into the CNS, driving inflammation and tissue injury [13, 15].

While CD4+ T cells have traditionally been the focus of MS research, recent studies highlight the importance of CD8+ T cells in disease pathogenesis. These cells can contribute to inflammation, tissue damage, and disease progression through various mechanisms, including the production of pro-inflammatory cytokines and the targeting of neural antigens [16–18].

In the last decade, numerous scientists have tried to find appropriate treatments for patients with MS [19, 20]. Although MS naturally occurs only in humans, it is not possible to usually perform direct manipulations in humans, so we rely on animal models mimicking the disease. Experimental autoimmune encephalomyelitis (EAE) is widely used as an animal model of MS [2].

Many pharmacological studies have been conducted to find new drugs for the treatment of MS; however, the current treatments are far from satisfactory. In recent years, herbal compounds have received substantial attention for the treatment of inflammatory and neurological diseases. Acetyl-11-keto-beta-boswellic acid is a natural triterpenoid derived from the resin of *Boswellia* species. This compound has demonstrated significant therapeutic potential across various disease models, attributed to its anti-inflammatory, antioxidant, endothelial, neuroprotective, and vasoprotective characteristics [21]. In recent years, research has focused on AKBA's effects in autoimmune and inflammatory conditions. Numerous studies have highlighted its therapeutic advantages in animal models of ischemic brain injury, dermatitis, and multiple sclerosis [22–25]. Notably, AKBA has been shown to aid in the repair of sciatic nerve injuries in

rats by enhancing the phosphorylation of the ERK signaling pathway. This process promotes the proliferation and myelination of Schwann cells, which are crucial for nerve regeneration [26]. Additionally, AKBA has exhibited beneficial effects in other disease models, including diabetes, inflammatory bowel disease, and central nervous system disorders resulting from ischemia–reperfusion injury [24]. AKBA has been found to reduce oxidative stress, inflammation, complement activation, and cell death in brain endothelial cells subjected to oxygen–glucose deprivation and subsequent reperfusion [27]. Despite the availability of current treatments for MS, many remain inadequate, prompting ongoing efforts to discover new therapeutic options. The encouraging findings from studies examining AKBA's impact on autoimmune and inflammatory diseases, along with its neuroprotective and vasoprotective effects, position it as a promising candidate for further investigation as a potential treatment for MS and other neurological disorders. However, the immunomodulatory activity and neuroprotection potentials of AKBA for the treatment of MS have not been thoroughly investigated. Thus, the aim of the current study was to investigate the effects of AKBA on immunological and neurological functions in a mouse model of multiple sclerosis.

Material and Methods

Animals

Female C57BL/6J (10-weeks old) mice were housed under Specific Pathogen Free (SPF) conditions. All experiments were carried out according to National Institutes of Health guidelines for laboratory animals and the Ethics Committee of Shahid Beheshti University of Medical Sciences, Tehran, Iran (Ethical Code: IR.SBMU.MSP.REC.1395.450).

Induction and Clinical Grading of EAE

EAE mice were obtained from Salari Institute of Cognitive and Behavioral Disorders (SICBD; Alborz, Karaj, Iran). Mice were individually weighed and anesthetized with intraperitoneal (i.p.) injection of 50 mg/kg ketamine and 5 mg/kg xylazine. Myelin oligodendrocyte glycoprotein peptide 35–55 (300 μ g in saline; MOG 35–55; KJ Ross-Petersen ApS, Denmark) was combined with an equal volume of complete Freund's adjuvant (CFA; Sigma, USA) containing 500 μ g of heat-killed mycobacterium tuberculosis. Then, this emulsion was injected subcutaneously at the base of the tail of each mouse. The mice also received i.p. injection of pertussis toxin (300 ng in 100 μ l PBS; List Biological Lab, USA) [28, 29]. EAE disease was graded on a 0–5 scale as follows: 0, 0.5, 1, 2, 3, 4 and 5 for observation of normal, partial tail weakness, complete tail weakness,

forelimbs paralysis, hind limbs paralysis, paralysis of two limbs and death, respectively. Additionally, mice weights were recorded daily.

Experimental Design

Animals were divided into three groups (6 mice per group; $n = 6$), namely, EAE, EAE + Vehicle, and EAE + AKBA. The EAE group did not receive any treatments until the end of the experiment on day 28. The EAE + Vehicle and EAE + AKBA groups received intraperitoneal injections of vehicle (DMSO; Dimethyl sulfoxide) or AKBA (100 mg/kg/day) respectively, starting from day 14 to day 28, following the onset of clinical signs. AKBA with a purity of 98% was purchased from Chromadex (Irvine, CA 92618, USA). The final concentration of DMSO was consistently maintained at 2.5% v/v throughout all of the experiments [30]. Clinical evaluation of EAE and animal weights were recorded daily before and after of the immunization. The animals were sacrificed at day 28 and both sera samples and spinal cord tissues were collected for future evaluation. Each experiment was repeated, and assays were performed in duplicate.

Cytokine Measurements

The concentrations of serum cytokines, including IFN- γ , TNF- α , IL-2, IL-10, IL-17A, IL-6, IL-1 β , IL-1 α , PGE2, MIP-1 and MCP-1 were measured using enzyme-linked immunosorbent assay kits purchased from Abcam and R&D System. Assays were performed in duplicate.

Histopathological Assessments

Twenty-eight days after EAE induction, mice were sacrificed followed by perfusion-fixation with 10% formaldehyde. In brief, the spinal cord tissue was obtained and fixed in formaldehyde 10% for paraffin embedding. Then, the specimens were cut at 5 μ m and stained with Luxol fast blue (LFB) as well as hematoxylin and eosin (H&E) to assess myelin and quantify cell infiltration. The stained tissue samples were subsequently examined. To evaluate and analysis of the histopathological changes, ImageJ™ software was used.

Western Blot Analysis

To isolate crude protein, lumbar sections were dissected from the spinal cord tissue and digested with lysis buffer (Thermo Fisher Scientific) and centrifuged (12,000 RPM for 15 min at 4°C). Then the total protein concentration was measured using standard Bradford method. After that, crude proteins were isolated using SDS-PAGE electrophoresis followed by transferring onto poly vinylidene difluoride (PVDF) membrane. Then, the membrane being blocked with

5% Bovine serum albumin (BSA) at room temperature (RT) for 2 h followed by overnight incubation with primary antibodies against the allograft inflammatory factor 1 (Iba-1), glial fibrillary acidic protein (GFAP), myelin basic protein (MBP) and oligodendrocyte transcription factor-2 (Olig-2) at 4 °C. At next stage, the membrane was washed with Tris-buffer saline, containing 0.1% Tween 20, and incubated with secondary antibodies (Santa Cruz, Germany) at ambient temperature under continuous shaking. At final step, the specific bands were visualized via ECL (Enhanced Chemiluminescence) assay (Fermentas, Germany), and each protein expression was normalized to β -actin level. Regarding to assessment of the immunoreactivity, ImageJ™ software was used to measure signals.

Statistical Analysis

Statistical analysis was performed using GraphPad Prism 5.0. The clinical severity of EAE and weight changes in different time point were analyzed by Repeated measures two-way ANOVA. The serum cytokine levels in the EAE and EAE + AKBA groups were assessed using *t* test. To evaluate the histopathological changes and the expression of proteins between groups, One-way ANOVA were performed. All data were shown as mean $\pm$ SD. $p < 0.05$ was considered statistically significant.

Results

AKBA Improved the Clinical Severity of EAE

As shown in Fig. 1A, AKBA-treated mice exhibited significantly reduced disease severity compared with EAE + vehicle and EAE groups ($P < 0.5$). Our results showed that the average body weights of EAE mice had substantially declined ten days post-immunization. On the other hand, the average body weights of EAE mice were significantly increased after administration of AKBA (Fig. 1B; $F = 7.058$, $P < 0.05$).

AKBA Modulated Serum Cytokines and Chemokines Levels in EAE mice

ELISA assay for the serum cytokine/inflammatory mediators (IFN- γ , TNF- α , IL-2, IL-10, IL-17A, IL-1 β , IL-6, IL-1 α , and PGE2) and chemotactic cytokines (MIP-1 and MCP-1) were performed. A significant decrease was observed in the levels of pro-inflammatory cytokine TNF- α (13.27 ± 3.42 (EAE group) vs. 3.28 ± 1.47 (EAE + AKBA) pg/ml, 4.04-fold; $p < 0.001$) in EAE + AKBA mice compared to the EAE group (Fig. 2B). The mean concentrations of IFN- γ (740.1 ± 94.81 vs. 1119 ± 43.31 pg/ml, 1.51-fold;

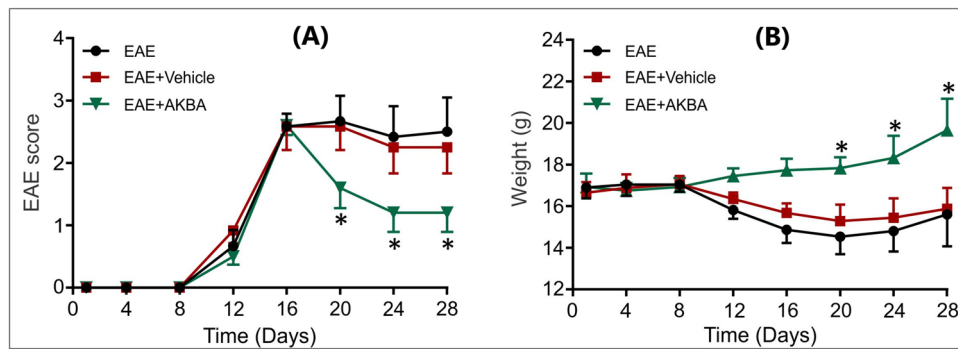
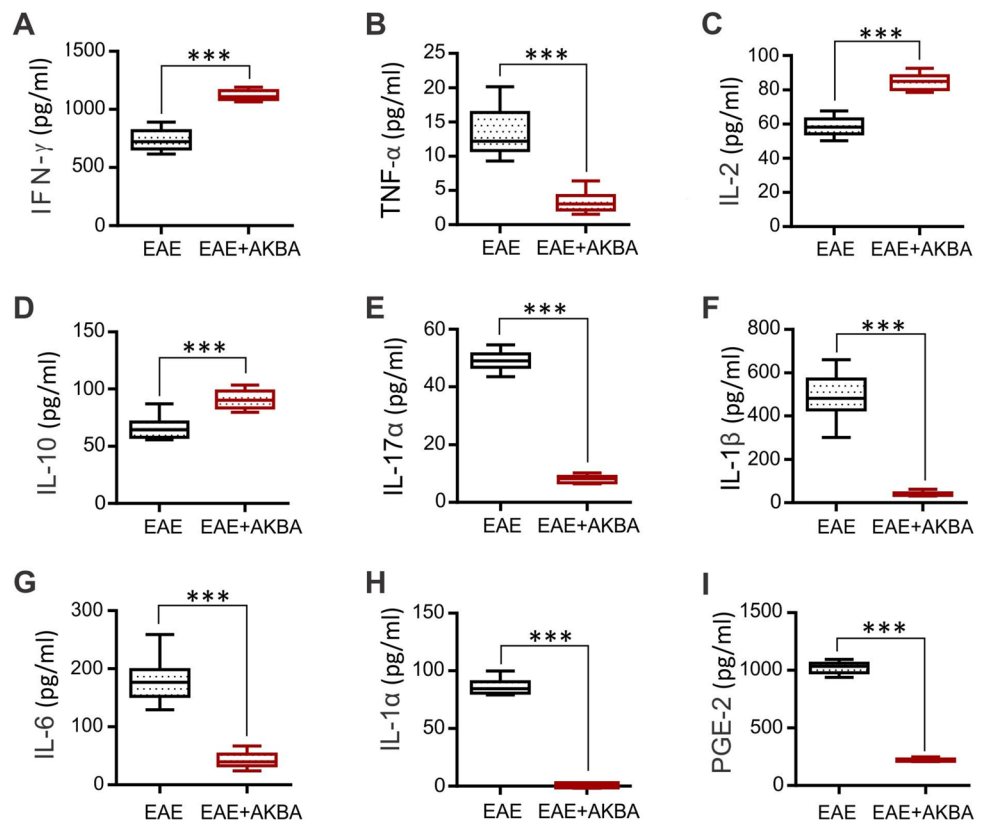


Fig. 1 The effects of AKBA on the clinical scores and averaged body weight in EAE Mice. (A) AKBA administration in EAE mice significantly reduced the clinical severity score at the specified time points compared with the EAE and EAE+ vehicle groups. (B) Mice treated

with AKBA showed a significant increase in body weight compared with the EAE and EAE+ vehicle groups. Results are presented as mean $\pm$ SD ($n=6$ /group). Asterisks (*) indicates $P<0.05$ vs. control groups

Fig. 2 The effect of AKBA on serum concentrations of inflammatory cytokines/mediators in EAE mice. The mean levels of Th1 cytokines are shown for EAE and EAE + AKBA groups. (A), IFN- γ , (B) shows TNF- α , and (C) represents IL-2. The mice that received AKBA showed a significant decrease in pro-inflammatory cytokine TNF- α and a significant increase in IFN- γ and IL-2 levels. (D) AKBA administration significantly elevated the anti-inflammatory cytokine IL-10 in EAE mice compared with the EAE + AKBA group. (E) Treatment with AKBA reduced the mean value of pro-inflammatory cytokine IL-17A compared to EAE mice significantly. The mean levels of peripheral monokines IL-1 β (F), IL-6 (G), IL-1 α (H), and PGE2 (I) decreased significantly compared with the EAE group. Results are presented as the mean $\pm$ SD, ($n=6$ /group). ***: $p<0.001$, denotes significance



$p<0.001$) and IL-2 (58.70 ± 5.60 vs. 84.99 ± 4.88 pg/ml, 1.44-fold; $p<0.001$) were significantly higher in EAE + AKBA mice relative to the EAE mice (Fig. 2A and 2C). We observed higher levels of anti-inflammatory cytokine IL-10 in EAE + AKBA mice compared with the EAE group (66.94 ± 10.20 vs. 90.66 ± 8.34 pg/ml, 1.35-fold; $p<0.001$, Fig. 2D). There was also a significant difference between the mean values of pro-inflammatory Th17 cytokine IL-17A in EAE + AKBA (8.12 ± 1.23 pg/ml) and EAE mice (49.07 ± 3.24 pg/ml), which indicates

that the serum level of IL-17A was 6.04-fold higher in the EAE group compared with EAE + AKBA group (Fig. 2E; $p<0.001$). A significant decrease was also observed in the levels of peripheral monokines IL-1 β , IL-6, IL-1 α and PGE2 in EAE + AKBA mice in comparison with the EAE group; IL-1 β (498.60 ± 98.96 vs. 41.43 ± 9.09 pg/ml, 12.03-fold; $p<0.001$, Fig. 2F), IL-6 (180.20 ± 36.42 vs. 41.76 ± 10.51 pg/ml, 4.31-fold; $p<0.001$, Fig. 2G), IL-1 α (86.36 ± 6.56 vs. 0.72 ± 0.24 pg/ml, 119.94-fold; $p<0.001$, Fig. 2H) and PGE2 (1024 ± 74.72 vs. 224.4 ± 12.28 pg/ml, 4.56-fold;

$p < 0.001$, Fig. 2I). There was also a significant difference between the mean values of chemotactic cytokines MIP-1 (89.13 ± 5.09 vs. 24.44 ± 1.82 pg/ml, 3.64-fold; $P < 0.001$, Fig. 3A) and MCP-1 (3132 ± 220.6 vs. 1717 ± 85.87 pg/ml, 1.82-fold; $P < 0.001$, Fig. 3B) in EAE + AKBA mice and EAE mice.

AKBA Regulated Myelination, Cells Infiltration and Gliosis in EAE

LFB staining was performed to determine the percentage of demyelination area in microscopic sections. Our analysis of the LFB staining results revealed that the administration of AKBA in EAE mice decreased the area of demyelination in this group compared with the other groups (Fig. 4A). H&E

staining results are shown that the mean number of inflammatory cell infiltration had a significant reduction in the spinal cord of EAE + AKBA mice group (Fig. 4B; $p < 0.05$). WB was performed to examine the effects of AKBA on the myelination and gliosis in all experimental groups (Fig. 5A1-A3 and Fig. 5)B1-3. Blotting results showed that the expression of GFAP significantly decreased in the EAE + AKBA group compared with that of EAE + vehicle and EAE groups (Fig. 5A-2; $p = 0.001$). The value of Iba1 expression in EAE + AKBA group demonstrated also a significantly decline (Fig. 5A-3; $p = 0.001$). In contrast to the EAE and EAE + AKBA groups, the level of MBP in the EAE + AKBA group significantly increased compared with the other experimental groups (Fig. 5B-2; $p = 0.006$). A significant increase was also observed in the expression

Fig. 3 The expression of chemotactic cytokines regulated with AKBA administration in EAE mice. (A and B) After injection of AKBA in EAE mice, the levels of chemotactic cytokines, namely, MIP-1 (A) and MCP-1 (B) were significantly lower in EAE + AKBA group compared with EAE mice. Results are presented as the mean $\pm$ SD, ($n = 6$ /group). ***, $p < 0.001$, denotes significance

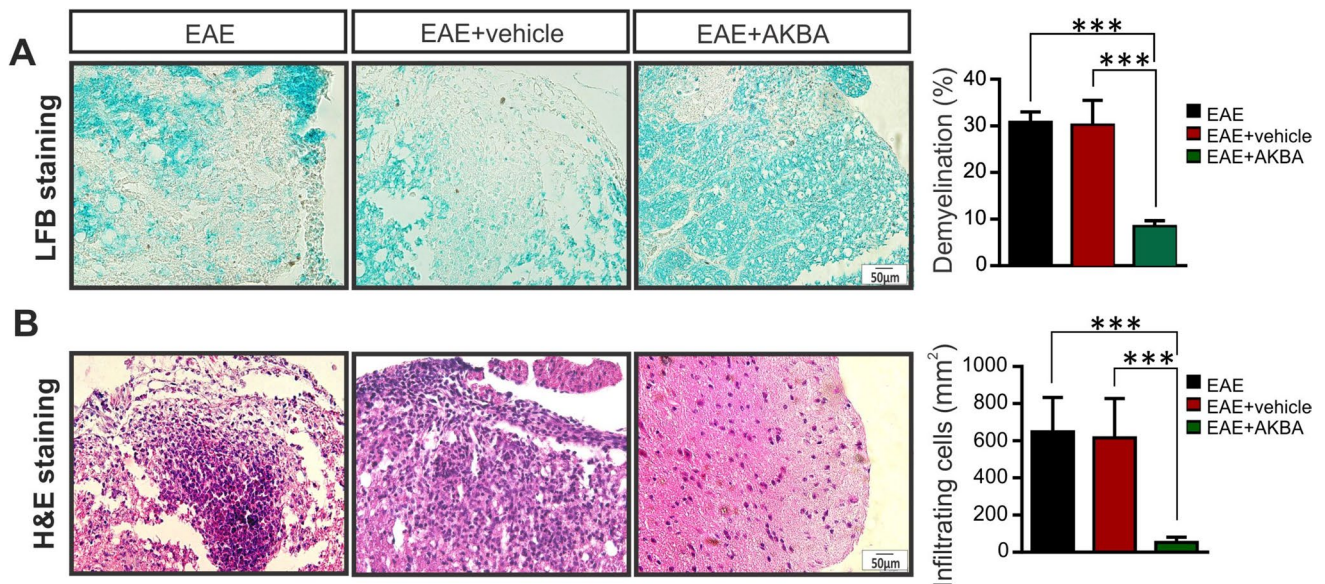
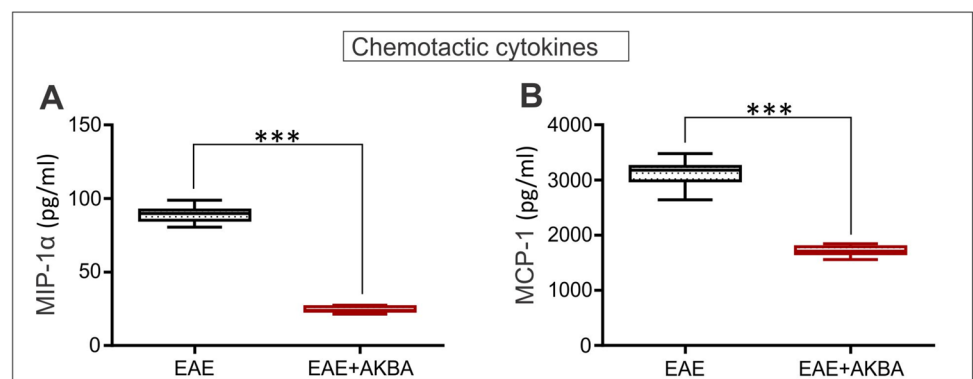


Fig. 4 Luxol fast blue (LFB) and hematoxylin and eosin (H&E) staining are used to evaluate histopathological changes of the lumbar sections dissected from the spinal cord in different experimental groups. (A) Quantitative data analysis of LFB staining in all experimental groups vividly show that the collective percentage of demyelination areas in spinal cord of EAE mice treated with AKBA were significantly

decreased. (B) The number of inflammatory cells infiltration in spinal cord also was significantly lower in EAE + AKBA group in comparison to EAE mice as confirmed by the H&E staining. Results are presented as the mean $\pm$ SD, ($n = 6$ /group). ***, shows significance at $p < 0.001$ level

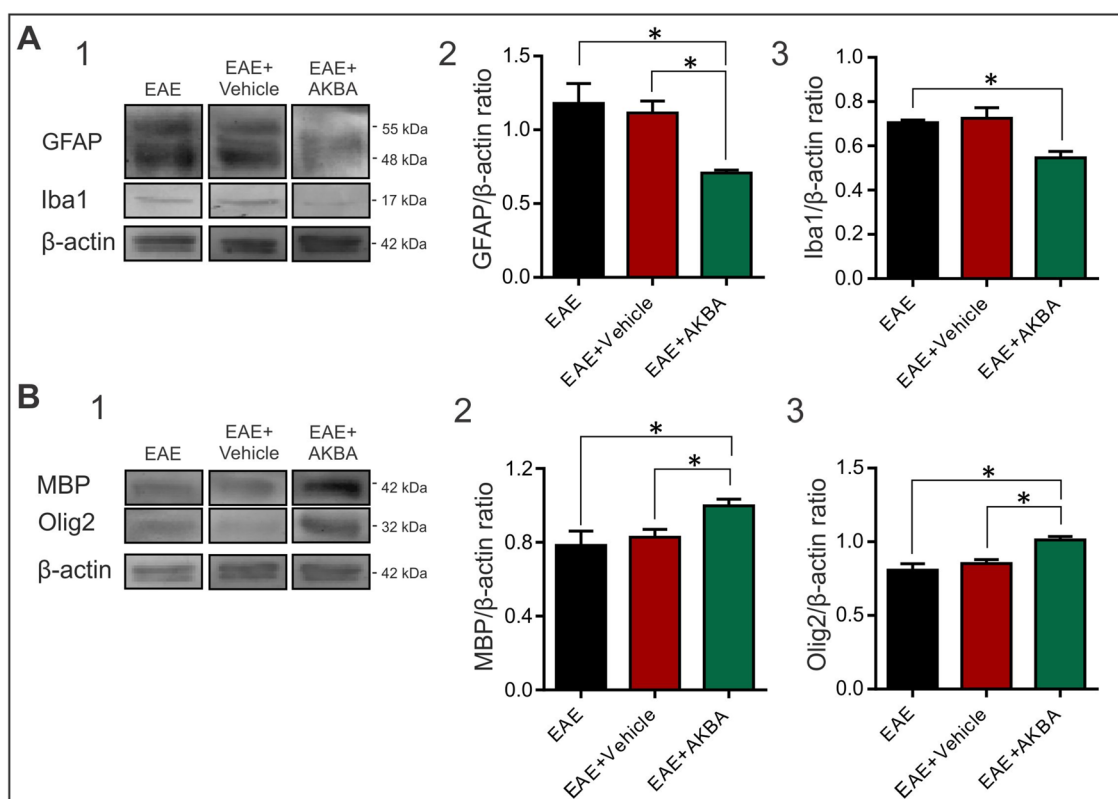


Fig. 5 Western blot analysis of GFAP, MBP, Olig-2, Iba-1 protein expression in spinal cord of EAE mice with and without AKBA treatment (A) The protein levels of GFAP and Iba-1 significantly declined in the AKBA-administered EAE mice compared with EAE and EAE+ vehicle groups. [1] shows the specific bands on the PVDF membrane, whereas and depicts the normalized ratios for GFAP and Iba-1, respectively. (B) Treatment of EAE mice with AKBA

elevated the expression of MBP and Olig2 compared with the EAE and EAE+vehicle groups [1], visualizes the bands, and, represent the normalized ratios for MBP and Olig2, respectively. Levels of each protein in both cases, (A and B), are normalized to that of β-actin expression. ImageJ was used to quantify signals and assess the immunoreactivity. Results are presented as mean ± SD, (n = 3/group). “***”, shows significance at $p < 0.001$ level

of Olig2 in the AKBA-treated EAE mice compared with the other groups (Fig. 5B-2; $p < 0.001$).

Discussion

Our findings revealed that AKBA treatment in EAE mice significantly decreased the EAE severity, demyelination, inflammatory cells infiltration, gliosis and as well as significant changes in the expression of serum inflammatory cytokines and markers such as IFN- γ , TNF- α , IL-2, IL-10, IL-17A, IL-6, IL-1 β , IL-1 α , PGE2, MIP-1, and MCP-1 were observed.

MS is an inflammatory demyelinating disorder of CNS. The current treatments for MS are not satisfactory in its progressive stage [31]. Today, scientists focus on remyelinating and immunomodulatory therapies for MS [32, 33]. Since herbal compounds could improve myelin repair and suppress inflammation, there is a desire for herbal-based therapy for

MS with the purpose of promoting CNS regeneration [34, 35].

According to the previous studies, the therapeutic effects of AKBA have been shown in several animal models with neurological disorders such as ischemic brain injury [36], Parkinson’s disease [37] and middle cerebral artery occlusion [38]. Nevertheless, the therapeutic effects of AKBA for the treatment of MS have not been thoroughly studied so far. Our results revealed the therapeutic effects of AKBA in the treatment of EAE mice were in total congruence with previous research. It was previously reported that AKBA attenuated the neurological deficits in ischemic brain injury [36], our results also demonstrated that AKBA attenuated the EAE clinical signs. The protective effects of AKBA on myelination in EAE mice could be presumably attributed to the higher levels of MBP and Olig2. Furthermore, LFB staining also confirmed that AKBA modulated myelination. These results are concurrent with those formerly observed in the case of curcumin, that alleviated EAE symptoms via promoting neuroprotection and myelination [8].

Several investigations have pointed out the immunomodulatory effects of AKBA in different diseases [39]. Several studies demonstrated that IL-17A-producing Th17 cells play a dominant role in the EAE induction [40, 41]. Furthermore, it has also been reported that AKBA decreases IL-17A production slightly by blocking the differentiation of Th17. Additionally, AKBA facilitates the Th2 and Treg cells differentiation without a distinct effect on the differentiation of Th1 cells [42]. In confirmation of the previous data this study showed that AKBA as an anti-inflammatory agent significantly downregulated the levels of Th1 cytokine TNF- α , Th17 cytokine IL-17A, peripheral monokines IL-1 β , IL-6, IL-1 α and PGE2, as well as chemotactic cytokines MIP-1 and MCP-1 in the EAE mice. On the other hand, significant increases of IL-2, IFN - γ , and IL-10 were observed in EAE + AKBA mice compared with the control groups. In addition, H&E staining indicated that treatment of EAE mice with AKBA attenuated the immune cell infiltration into spinal cord. Moreover, Shehata et al. showed that AKBA can decrease the levels of TNF- α , IL-6, IL-1 β , and IL-1 α inflammatory cytokines in animal model of diabetes [43]. It was also reported that *B. serrata*'s gum resin administration in rheumatoid arthritis significantly declined the concentrations of IL-1 β , TNF- α and IL-6 pro-inflammatory cytokines, and increased IL-10 anti-inflammatory cytokine [44]. In addition, a previous work by Ammon et al. demonstrated that AKBA by inhibition of nuclear factor-kappa B can trigger a decrease in the levels of TNF- α and interleukins [39]. In harmony with our findings, an in vitro study has also indicated that AKBA suppresses the production of PGE2 and thereby further suppresses Th17-cell differentiation [45, 46].

MIP-1 and MCP-1 play a crucial regulatory role in T helper lymphocyte differentiation and recruitment of leukocytes at sites of tissue inflammation [47]. According to the previous studies and our results, it could be suggested that AKBA, plays a dominant role in reduction of inflammatory cells infiltration into the spinal cord of EAE mice via suppressing MIP-1 and MCP-1 chemokines expression.

Current treatments of MS are costly and have various side effects. According to our results, it could be proposed that AKBA offers numerous advantages over the previous and contemporary treatments. Recent studies on herbal compounds such as AKBA have indicated numerous benefits, particularly immunomodulatory effects, safety, and widespread availability [39]. Researchers have kept herbal-based therapies for MS under the radar with the aim of promoting CNS neuroprotection [48] and the current study demonstrates the promising therapeutic effects of AKBA in ameliorating inflammation and demyelination encountered in MS. Alleviating the severity of EAE symptoms using AKBA and the resultant neuroprotection, remyelination and immunomodulation, as well as suppression of gliosis

and infiltration offered by it might pave the way to novel concomitant/adjuvant MS therapies that well worth further investigations. In our study, significant reductions in pro-inflammatory cytokines (TNF- α , IL-17A, IL-1 β , IL-6, IL-1 α , PGE2) and increases in anti-inflammatory cytokines (IL-10) in the serum of AKBA-treated mice suggest an immunomodulatory effect on peripheral immune cells. Additionally, decreased levels of MIP-1 and MCP-1 in AKBA-treated mice indicate a potential impact on the migration of immune cells toward the central nervous system. Moreover, reduced demyelination, inflammatory cell infiltration, and gliosis in the spinal cord of AKBA-treated mice suggest a direct neuroprotective effect within the CNS. The increased expression of MBP and Olig2 in AKBA-treated mice indicates enhanced remyelination and oligodendrocyte survival. While the current study provides evidence for both peripheral and central effects of AKBA in EAE mice, further research is needed to fully elucidate its mechanism of action. Tregs play a crucial role in maintaining immune homeostasis and preventing excessive inflammatory responses. Unfortunately, due to the scope of the current study, we did not include Treg data in our analysis. We recognize this as a shortcoming and plan to address it in future studies. Moving forward, we will make a concerted effort to incorporate Treg data to provide a more comprehensive understanding of the immunomodulatory effects observed in our research. Investigating the questions outlined above would significantly contribute to our understanding of how AKBA exerts its therapeutic effects and potentially inform the development of more targeted therapies for MS.

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Author Contributions All authors contributed to the study's conception and design. Material preparation was done by Farzaneh Salmani, Sara Khoshtinatnikhouy and Reza Ataei Kachouei. Data collection and analysis were performed by Ali Jahanbazi Jahan-Abad, Abbas Tafakhori, Sajad Ehtiati, Seyyed Hossein Khatami, and Hadis Firoozpour. The first draft of the manuscript was written by Saeed Karima and Seyyed Hossein Khatami. All authors read and approved the final manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethics Approval All experiments were approved by the ethics committee at Shahid Beheshti University of medical sciences (Ethical Code: IR.SBMU.MSP.REC.1395.450) and were conducted in accordance with Animal Care Committee guidelines of Shahid Beheshti University of medical sciences.

Conflict of Interest The authors declare no competing interests.

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